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AI Trust-Score for Online Content.

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***Abstract*—The massive implementation of generative Artificial Intelligence (AI) in educational systems, news, healthcare support, governance procedures, customer support, and source code engineering has rendered trust assessment a core issue. These days, most systems display either unproven results or meaningless confidence metrics that aren't meant to support human decision-making. In this paper, AI Trust Score System (AITSS) will also be presented, a convenient and understandable model of predicting the credibility of AI-created content based on a single trust score and factor-diagnosis. The proposed architecture uses a weighted scoring system that takes into account danger indicators, unclear behavior, consistency of assertions, source reliability, and explainability quality. An implementation of real-time scoring and retrospective auditing to facilitate persistent storage and dashboard-level interpretation is created based on Spring Boot. Mixed-quality text samples provide clear evidence of score separation across trust categories among high-trust, moderate-trust, and low-trust content classes. Additionally, the study shows that open component breakdown improves actionability and human reviewer agreement compared to scalar confidence-only interfaces. The framework is policy driven and modular to allow domain specific calibration of sensitive apps. The future work involves multimodal extension, adaptive weight learning in relation to expert feedback, and sound benchmark-based calibration**

***Index Terms*—AI trust, reliability scoring, confidence calibration, content verification, risk-aware AI**

I. Introduction

Generative Artificial Intelligence (AI) has come out of being an experimental demonstration to one of the pillars of modern digital infrastructure. Currently, organizations are delegating the AI systems with report writing, summarizing legal and policy documents, writing software code, offering educational descriptions, supporting customers and synthesizing multi-modal content. Many sectors, including those developed by OpenAI and Google DeepMind, have seen a notable gain in productivity due to the quick operationalization of huge language models. However, this shift has shown a deep divide in the production of output and the accuracy of the same output. Practically speaking, even tenuously fact-grounded and fabricated-reference response chains with unstable reasoning chains can be generated using generative AI systems in a manner that sounds persuasive and fluent. Hallucination, fabricated citations, and factual inconsistency have been cited before as the continuing trouble with large language models [1], [2]. In the high-impact utilization of linguistic fluency is not a proxy of trustworthiness. This leads to the fact that the difference between the surface coherence and the epistemic reliability becomes the central issue in applied applications. The issue of trust is especially consequential since users are already required to engage with AI systems in time constraints on a regular basis. Decision-makers may be bandwidth-limited to make sure that all the statements they generate are checked manually. Optimized AI answers can be considered true by the students. Analysts are unaware that they are basing their conclusions on flimsy summaries. Teams can use the outputs of AI in the downstream automation pipelines where mistakes spread and multiply. Under these conditions, the lack of ex-plicit transporting mechanisms of trust shifts the responsibility of verification to the people completely and thus creating the operational risk and unreliable quality control. [3] The binary choice paradigm, in which the outcome is either accepted or rejected, is frequently assumed in conventional system design. Real AI workflows require more than just a binary approach. It is best to think about reliability as a continuum. Reliability is viewed as a continuum. There are extremely reliable and reusable outputs, which need few or no reviews; there are partially useful yet verifiable ones, and they are unsafe and must be restricted or prioritized. A graded representation of trust, in this regard, is essential to subtle, policy-congruent decision-making. The second weakness that is permeating most deployments is the area of opacity. A model that displays a value that resembles confidence is under-explained and model-specific even in instances where a model exposes a value that resembles confidence. The users cannot determine whether confidence is a factual correctness, probabilistic certainty or a heuristic scoring. Without interpretability, trust signals cannot be audited, and they cannot be improved easily. In enterprise and governance contexts, this kind of opacity leads to accountability issues, with teams not able to explain why something was accepted, flagged or rejected in a content item [4]. A third challenge is the issue of variability. AI's output can differ depending on the model version, context window, prompt formulations, and decoding parameters. The research conducted in model calibration and stability of responses proves that the confidence of the neural network is often out of touch with actual correctness [5]. It is possible that

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consistency in a response on first presentation is not consistent when repeated later. This variability compromises the use of a single instance of output; therefore, a confidence result should be based on a measurement of uncertainty behaviors and consistency patterns between similar checks.

With such dilemmas, there is an urgent need to revise the concept of trust as an abstract axiom into something that is quantifiable, explainable, and configurable as a system component. The current article attempts to meet this need by creating Artificial Intelligence Trust Score System (AITSS) which is based on three practical goals. First, the framework provides a scalar trust score that is within a normalized range (0100), thus making it possible to triage and make interventions based on policy. Second, it disaggregates to a factor wise level and retains interpretability and enables a structure of human review. Third, it supports long-term storage and trend analysis and therefore makes it possible to track it longitudinally, instead of scoring it ad hoc.

Instead of a monolithic trust outcome, the proposed architecture aims to synthesize a variety of visible indicators that touch upon various aspects of reliability, that is, the credibility of a source, consistency of claims, uncertainty behavior, the quality of explainability, and risk measures. Multi-factor trust modelling is consistent with modern AI governance paradigms of focusing on transparency, accountability, and risk-based control [6] The solution facilitates rigorous assessment by combining algorithmic transparency with production-ready software engineering to make the solution useful in both research and in practice. A real-world implementation case served as the inspiration for this paper: a web-enabled trust assessment system built using the Spring Framework (Spring Boot), storing data in a JPA-compliant storage system, and providing score and monitoring data via dashboard/API endpoints. Engineering orientation is intentional, which is an indication that goes beyond the conceptual modelling to provide an operational architecture that can be operationalized in the real enterprise workflows. AITSS provides a systematic channel to AI trust management in operation through the combination of computations of trust, explicable diagnostics, and continuous audit trails

II. Literature review

The rapid expansion of artificial intelligence and social media has intensified concerns about misinformation, fake news, and AI-generated content. Researchers have therefore focused on detection mechanisms, forensic analysis, and watermarking to counter these threats. This review synthesizes key findings from recent surveys and studies on fake news detection, AI-generated text identification, and deepfake detection. Fake news detection is a major challenge in the digital era. According to [7], fake news is intentionally misleading or false information disseminated through online platforms. Detection methods are typically grouped into content-based, context-based, and hybrid approaches. Content-based methods emphasize linguistic cues such as syntax, semantics, and sentiment, while context-based methods examine user behavior, social interactions, and propagation patterns. Hy- brid approaches combining both are generally more effective. Nonetheless, persistent challenges include limited data, adversa rial manipulation, and rapidly evolving misinformation strategies. With the rise of large language models (LLMs), identifying AI-generated text has become increasingly important. Study [8] reviews detection strategies ranging from statistical analysis and traditional machine-learning classifiers to deep neural networks. It notes that AI-generated text often closely resembles human writing, making detection difficult. Adversarial tactics such as paraphrasing and prompt engineering further reduce accuracy, indicating the need for more robust and adaptive methods. Expanding this perspective, [7] Analyses AI-generated text forensics across three components: detection, attribution, and characterization. Detection determines whether content is AI-generated, attribution attempts to identify the source model, and characterization examines stylistic patterns. The study suggests that combining techniques like stylometry, metadata analysis, and probabilistic modeling can improve results. However, scalability and the lack of standardized evaluation benchmarks remain significant constraints. Watermarking has emerged as a promising approach for source-level identification of AI-generated content. In [10], a probabilistic watermarking method for LLMs embeds identifiable patterns into generated text while preserving quality. This enables reliable detection without undermining usability. Complementing this, [11] surveys watermarking techniques and categorizes them into statistical, cryptographic, and hybrid approaches. Despite their poten- tial, watermarking methods remain vulnerable to text editing and paraphrasing and raise ethical concerns regarding transparency and user consent. Deepfake detection is another critical domain given advances in generative models for images, video, and audio. Study [12] surveys datasets, algorithms, and challenges, classifying detection techniques as spatial, temporal, and frequency based. Spatial methods detect inconsistencies within frames, temporal methods analyze motion patterns across frames, and frequency-based methods detect artifacts in the frequency domain. Key limitations include weak generalization across datasets and sensitivity to high-quality deepfakes. Similarly, [13] focuses on deep-fake detection for human face images and videos, covering both traditional machine-learning and deep-learning approaches such as CNNs and RNNs. The study highlights the need for large, annotated datasets and robust feature extraction. However, the rapid progress in deepfake generation continues to outpace detection methods. In audio deepfakes, [14] surveys speech deepfake detection using feature-based, model-based, and hybrid approaches. Feature-based methods analyze acoustic signals such as pitch and spectral characteristics, while model-based methods employ deep learning classifiers. The study emphasissizes that multimodal systems combining audio, visual, and textual information can substantially improve accuracy.Re-

maining challenges include cross-lingual detection, real-time performance, and robustness to adversarial attacks. Recent work underscores the growing need to detect AI-generated multimedia and ensure authenticity. Survey [15] provides a comprehensive overview of detection methods for text, image, audio, and video generated by large AI models, stressing the value of multimodal detection and strong benchmarks. Study [16] focuses on LLM-generated text detection, outlining statistical, machine-learning, and neural approaches while noting vulnerabilities to adversarial attacks and paraphrasing. In addition, [17] surveys text watermarking for LLMs, highlighting its role in source identification while recognizing limitations in robustness. Overall, the literature shows meaningful progress in detecting fake news, AI-generated text, and deepfakes, yet several challenges persist. These include the rapid evolution of generative models, limited standardized datasets, adversarial attacks, and scalability constraints. Future research should prioritize robust, multimodal, and explainable detection systems, while integrating watermarking and forensic techniques to improve reliability.

III. Problem Statement

In the field of research on artificial intelligence, we take an AI-generated object artifact x, and our main task is to evaluate a reliability measure T(x), which reflects the expected reliability of the said object in the stated policy environment and within the risk appetites outlined. Notably, the evaluation of trust is provided as a structured scale of reliability, instead of a dichotomous or binary result of validity, making it easier to make subtle, graded judgement in operational AI situations.

IV. Related Work

According to the current academic opinion, the ideology of artificial intelligence is spread across a wide spectrum of disciplinary areas, such as creating explainable AI and uncertainty prediction methods, identifying hallucinations, controlling AI safety, and the dynamics of human-AI interaction. The current methodologies tend to focus on these dimensions separately and not as a whole. Individually, examples of factuality verification systems can be designed without considering uncertainty-stability analysis; probabilistic measures of confidence can be designed without considering source credibility analysis; safety filters can be designed as standalone classifiers that do not give transparent factor-level explanations. Therefore, the existing deployments create disjointed reliability indicators, which are no longer coherent in their interpretation and policy resolution. The AI Trust Score System (AITSS) aims at closing this gap by combining several dimensions of trust into a single system of operation. The proposed system unites interpretability, uncertainty modelling, credibility assessment, and safety-wise scoring to provide a centralized and practical trust representation applicable in the deployment of AI to the real world.

V. Methodology

In this subsection, a systematic review of the major research lines of the forensic analysis of AI-generated content (AIGC) is provided, and the way in which these research lines apply to the AI Trust Score System is explained. The discussion that follows falls under the categories of text, image, video, audio, and multimodal content, representing the state of the art in technologies, datasets, and practical implications.

A. Detection AI-Generated Text & Images

Text Currently, transformer-based large language models (LLMs) are the dominant approach to text generation and include the families of GPT and LLaMA along with a variety of instruction-tuned models. These architectures can generate long-form prose that is coherent and more indistinguishable between human-written and machine-generated text, which makes it harder to detect the origin of the text in the context of misinformation in the real world. Technologies Current text-forensic systems use perplexity-based score, stylometric analysis, token- distribution fingerprints and supervised classifiers constructed with transformer encoders. Practical systems are increasingly incorporating model-based detection together with contextual metadata as well as provenance of sources to improve robustness.

Datasets: The most common corpora used in evaluations include HC3, GPT -WikiIntro, Turing Bench and domain-specific social/news collections. As a result, multilingual and code-mixed standards are limited, and this compromises the reliability in the multilingual user context.

Conclusion: Text-forensics is relatively mature but issues of cross domain generalization and resistance to paraphrase continue to exist. A mixed scoring system is thus proposed to be used in production systems.

Image

Image synthesis has developed since initial-generation GANs to the modern diffusion pipelines which can generate photorealistic fidelity with fine-tuning control of style to reduce the utility of legacy detectors.

Technologies: In most modern image-forensic methods, CNN or ViT-based detectors, frequency-domain artifact analysis, color/ noise consistency checks, and metadata-level verification are used. In the processes of identifying diffusion-era products, ensemble methods are normally required to provide adequate accuracy.

Datasets: The commonly used standard benchmarks are GenImage, CIFAKE and real- vs synthetic collections. Scholars also create customized data that compares real-life pictures with fabricated data of several engines to test the healthiness of detectors.

Conclusion: Diffusion models have brought visual realism to the point that single- signal detectors are inadequate. Good forensic systems thus combine artefact examination, metadata examination, and trained classifier results.

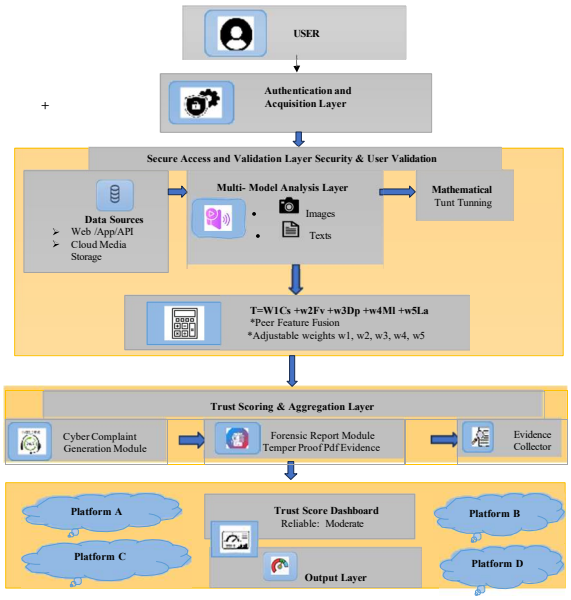


Fig. 1. Architecture of

The AI Trust Score System is a multi-layered architecture designed to evaluate the authenticity and reliability of digital content such as images, videos, audio, and text. It integrates advanced AI models, security mechanisms, and data processing techniques to detect misinformation and deepfake content. The system works through several interconnected layers, each performing a specific function from data collection to result presentation are shown in Fig.1. Overall, it ensures a transparent, secure, and accurate trust evaluation process.

User Layer

The User Layer is the main entry point where people and organizations interact with the AI Trust Score system. It offers web, mobile, and API access so users can upload, images, or text and receive authenticity results. The interface prioritizes a clean UX with clear upload flows, progress tracking, and status updates. Basic identity checks or login controls help ensure accountability and prevent misuse. This layer bridges human interaction with the AI backend in a simple, reliable way.

Authentication & Acquisition Layer

This layer acts as both a security gateway and a data intake pipeline. It verifies users through login, API keys, or access controls before accepting content. After validation, it collects data from apps, APIs, or storage systems, checks formats, and performs initial cleanup. Early preprocessing steps like frame extraction, audio separation, or text normalization also happen here. Metadata such as timestamps and source details are captured to support later trust analysis. From a security point of view, it supports login and token-based access. From an intake point of view, it validates content types, accepts URLs or files, and captures metadata that later influences trust scoring. This is also the stage where raw content is normalized into analysis-ready form.

Multi-Model Analysis Layer

This is the intelligence core where specialized AI models inspect each modality. CNN-based systems analyze images and deepfake artifacts. Transformer models evaluate text for misinformation or semantic inconsistencies. Different modalities require different types of analysis. A deepfake image needs visual artifact detection and a suspicious text message needs linguistic and claim-level evaluation.

Feature Fusion & Trust Score Calculation Layer

This layer combines the model outputs into a single trust score using weighted fusion. Signals like source credibility, fact verification, deepfake probability, metadata integrity, and linguistic authenticity are mathematically balanced. Weights can be adjusted based on context, such as journalism, legal review, or social media monitoring. The result is a unified, interpretable score that reflects overall authenticity.

Trust Scoring & Aggregation Layer

Here, the final score is stored, organized, and turned into structured reports. It can generate cyber-complaint summaries or forensic PDFs for investigations. Security safeguards like SHA-256 hashing and digital signatures ensure the reports are tamper-proof. The layer also aggregates results over time to support audits and case histories.

Output Layer

The Output Layer presents result in a human-friendly format. Users receive dashboards, labels like “Reliable” or “Suspicious,” and clear visual indicators. Integrations allow results to flow into external platforms for moderation, enterprise use, or law-enforcement workflows. This layer focuses on clarity, transparency, and actionable decision support.

Table 1 presents the layered architecture of the proposed system, in which each layer is designed to perform specific operations for secure data handling, intelligent analysis, and reliable trust evaluation. The architecture adopts a modular and scalable approach to ensure efficient processing of multi-modal data.

Layer	Main Purpose	Details
User Layer	Human and system interaction entry point	Web pages, dashboards, upload forms, history and detail views
Authentication & Acquisition Layer	Secure access and collect incoming content	Login, JWT, request validation, uploads, media link ingestion, preprocessing
Multi-Model Analysis Layer	Analyze each modality	Text analysis, source checks, media forensics, NLP signals
Feature Fusion and Trust Score Calculation Layer	Combine risk and authenticity signals	Weighted score blending, signal alignment, AI/fake probability estimation
Trust Scoring and Aggregation Layer	Store and report results	History records, forensic summaries, report generation, case tracking
Output Layer	Present final results	Dashboards, labels, detail pages, RAG display, workflow guidance

B. *Knowledge-Based Deepfake Detection*

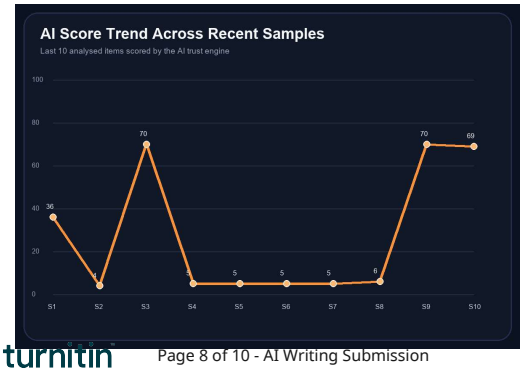
The highest concentration appears in the 0-30 band (True + Human), while a substantial number of samples also fall within the 70-100 band (Fake + AI). Fig. 2 presents the distribution of analyzed samples across four trust-score interpretation bands, offering a concise view of the scoring pattern observed in this study.

In contrast, the middle ranges, 30-50 and 50-70, contain comparatively fewer samples, indicating a clearer separation among the evaluated categories. This distribution supports the research paper's broader objective of examining how trust- score bands can help distinguish between human-generated and AI-influenced content.

Fig. 3 illustrates the variation in AI trust scores across the ten most recently analyzed samples, highlighting fluctuations in model assessment over time. The graph shows low scores for several samples in the middle sequence, followed by sharp peaks at S3, S9, and S10. This contrast indicates that the analyzed content does not follow a uniform trust pattern but instead reflects noticeable differences between samples. Such score variation is significant for the research paper, as it demonstrates how AI-based evaluation can capture changing reliability levels across different inputs.

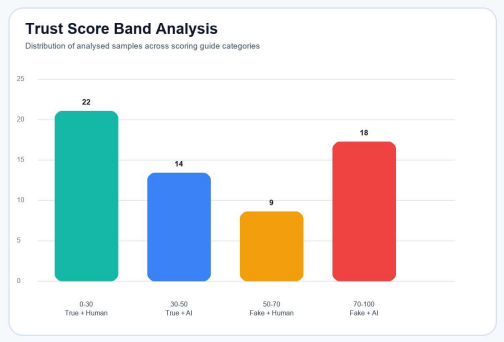
Fig. 5 presents the distribution of AI-detected and human-generated content through a donut chart, providing a clear visual summary of classification outcomes. The larger portion of the chart corresponds to content identified as not AI- generated, while a smaller segment represents highly AI- generated samples. This proportional difference suggests that most analyzed instances in the dataset align more closely with human authorship than strong AI generation. In the context of the research paper, this figure supports the comparative anal- ysis of authorship patterns and strengthens the interpretation of AI detection results.

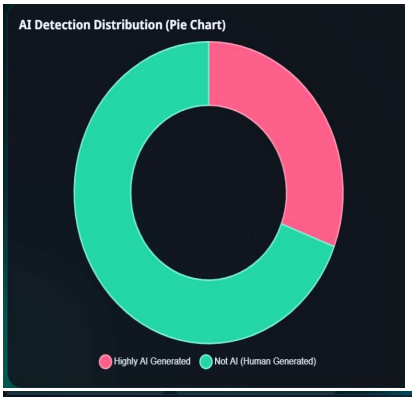
Fig. 6 presents the scoring guide used to interpret trust values by mapping score ranges to truthfulness and origin categories. Each band provides a structured basis for distinguishing between authentic human-written content, AI-assisted content, misleading human-generated content, and highly AI-generated fake content. This classification framework is important for the study because it standardizes how analyzed samples are evaluated and compared. In the research paper, the guide serves as a core foundation for explaining the trust-score based assessment methodology.

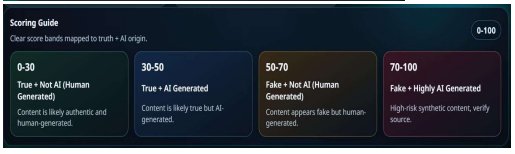


VI. Future Scope

The AI Trust Score System (AITSS) is a scholarly project that is moving towards evidence based, deployment ready, forensic intelligence platform. Future studies will focus on the multimodal reasoning integration to maintain semantic consistency on textual, visual, auditory, and audiovisual modalities. Introduction of platform-based metadata retrieval and safe archival facilitation will strengthen provenance checking procedures. The meticulously structured forensic reports will be produced with high-grade explainability practices, including the score dissections and institutional- specific integrity markers. Both strong governance systems including drift checks and calibration will be developed to ensure reliability in the long run. The code-mixed and multilingual upgrades to resilience will increase the applicability of the system in the context of heterogeneous lingual environments. In addition, real-time and edge-assisted modes of deployment will be included, which will allow rapid triage, all the way to facilitating more profound asynchronous forensic examination.







V. Conclusion

The AI Trust Score System has got a strong technical architecture and can develop into a high-impact forensic intelligence system. Most importantly, the future direction of the system must not be restricted to the simple increase in the number of model repositories, but rather aimed at developing understandable, explainable, and auditable multimodal reason- ing. The platform can address the burden of misinformation and synthetic media by implementing systematic benchmark governance, by introducing human-in-the-loop feedback, by enforcing strict deployment practices, and by ensuring privacy- conscious evidence management, which allows reducing the risks of misinformation and synthetic media in academic, institutional and operational settings.

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Feb. 2025